

JEREMIAH ADEBAYO

Agent Infrastructure & Machine Learning Engineer

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SUMMARY

IT student and ML/software engineer building autonomous agent systems and production-grade ML pipelines from first principles, including a multi-agent SWE automation system (LangGraph, hybrid retrieval, AST-based patching) and a from-scratch anomaly detection pipeline taken through quantization and edge deployment. Seeking remote roles at the intersection of agent infrastructure and applied ML.

PROJECTS

Noctis, Autonomous Multi-Agent Software Engineering System

May to June 2026

- Architected an LLM-based multi-agent code-fixing pipeline in LangGraph using Send-based parallel fan-out, scaling from single-file to concurrent multi-file bug resolution across a planner-to-engineer-to-critic loop with selective retry routing
- Built a hybrid retrieval (RAG) layer combining SQLite FTS5/BM25 lexical search with Voyage AI code embeddings (1536-dim) over Qdrant Cloud, feeding relevant repository context to a DeepSeek-based engineering agent
- Implemented AST-level code patching via libcst (vs. naive search/replace) for structurally reliable multi-file edits, orchestrated through a self-hosted LiteLLM proxy for multi-provider model routing (Cerebras, DeepSeek)
- Deployed and containerized the full pipeline (Docker sandboxed test execution) on cloud infrastructure

Nightfall, Multi-Category Visual Anomaly Detection (PatchCore, from scratch)

March to May 2026

- Reimplemented the PatchCore (Roth et al., 2021) computer-vision algorithm from the algorithm level in PyTorch (WideResNet50 multi-scale feature hooks, greedy k-center coresets selection, kNN memory-bank scoring), across all 15 MVTec AD categories, achieving 0.938 mean image AUROC (Area Under the ROC Curve) with full per-category AUROC/PRO evaluation
- Diagnosed a systemic INT8 (8-bit) quantization failure causing severe accuracy degradation across 8 MVTec categories (AUROC drops of 5 to 47 points, e.g., hazelnut 0.956 to 0.486, metal_nut 0.980 to 0.623); isolated two distinct root causes, a confidence-reweighting breakdown and a train/test feature-space mismatch, and resolved both via a config-gated fix and consistent-extractor memory-bank refit, recovering to 0.894 to 1.000 AUROC on validated categories
- Built and validated a full deployment pipeline including ONNX export (<1e-5 numerical deviation from native PyTorch), INT8 quantization (3.98x model-size reduction), and gRPC/REST serving, with real end-to-end inference validated against a Wokwi-simulated ESP32 client over a live network

EXPERIENCE

Software Engineering Intern

March 2026 to August 2026

UniK Connect

- Developed backend features in PHP for an inventory management system used by SME clients, covering authentication flows and database functionality supporting inventory, sales, and user workflows
- Configured and maintained deployment and server infrastructure for application availability; set up networking for SME businesses to run inventory and other business applications

Research & Data Analyst (Volunteer)

August 2025 to Present

OpenGov Africa

- Led data collection and Python/pandas analysis on Freedom of Information (FOI) law implementation across African countries within a self-directed research team, delivering a full report of findings

TECHNICAL SKILLS

Languages: Python, R, C++, JavaScript, SQL, PHP

Machine Learning & Deep Learning: PyTorch, TensorFlow, ONNX Runtime, XGBoost, LangGraph, LLMs / AI Agents, RAG & Vector Search (Qdrant), Voyage AI Embeddings, Computer Vision (anomaly detection)

Infrastructure & MLOps: Docker, gRPC, FastAPI, LiteLLM, SQLite/FTS5, Oracle Cloud (self-managed instances), model deployment & quantization (ONNX, INT8), REST API design

Other: Git, AST-based code manipulation (libcst), embedded systems (ESP32), pandas

EDUCATION

University of Ilorin (UNILORIN), Nigeria

Expected 2027

B.Sc. Information Technology, Third Year. **CGPA: 4.49 / 5.00**